HeatWise

USER FLOW DOCUMENT

Complete end-to-end user journey maps for the HeatWise rooftop cooling and garden planning platform — from first launch through scan, recommendation, visualization, and installation.

26	7	8
SCREENS	USER JOURNEYS	PAGES

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ALL USER JOURNEYS — OVERVIEW

1 ONBOARDING & AUTHENTICATION

Splash (launch) → Onboarding (walkthrough) → Phone Auth (OTP) → OTP Verify (confirm) → Profile Setup (name + city) → Home (dashboard)

2 CORE SCAN & MEASUREMENT FLOW

Home → Create Project (setup) → Photo Capture (camera) → AR Measurement (live/manual) → Environment (conditions) → Analysis (AI engine)

3 RECOMMENDATION & GARDEN LAYOUT

Analysis → Garden Layout (plan canvas) → Result (recommendations) → Report (full PDF)

Before / After (AI visual)

Install (quote)

4 EXPLORE & DISCOVERY (BOTTOM NAV)

City Heat (explore) → Species Library (browse plants) → Species Detail (deep info) → Create (start scan)

5 IMPACT & CARBON

Impact (dashboard) → Carbon (tracker)

↑ Carbon Setup (configure)

6 SETTINGS & UTILITY

Settings | Saved | Notifications | Tips | Install Done

BOTTOM NAVIGATION — GLOBAL ACCESS

Home	Explore	⬡ Scan	Impact	Settings
dashboard	city heat	primary CTA	dashboard	account

LEGEND: Core Screen · AI / Visual · Commerce · Utility

JOURNEY 1 — FIRST-TIME USER ONBOARDING

1 **Splash** → 2 **Onboarding** → 3 **Phone Number** → 4 **OTP Verify** → 5 **Profile Setup** → ✓ **Home**

Splash	Onboarding	Phone Number	OTP Verify	Profile Setup	Home
Logo animation 3 s auto-dismiss	3-card walkthrough Swipe or tap →	+91 number entry Sends SMS OTP	6-digit code 30 s resend timer	Name + City 2-step form	Dashboard Session active
AUTO	SWIPE	INPUT	VERIFY	FORM	DONE

↩ RETURNING USER — SESSION RESTORE

App Open (re-launch) → Session Check (NextAuth) → Home Dashboard (direct entry) or Phone Auth (expired session)

PROFILE SETUP — 2-STEP FORM DETAIL

STEP 1 — YOU

Display Name

Enter full name (min 2 chars)
Validates: non-empty string
Proceeds on "Continue →"

STEP 2 — LOCATION

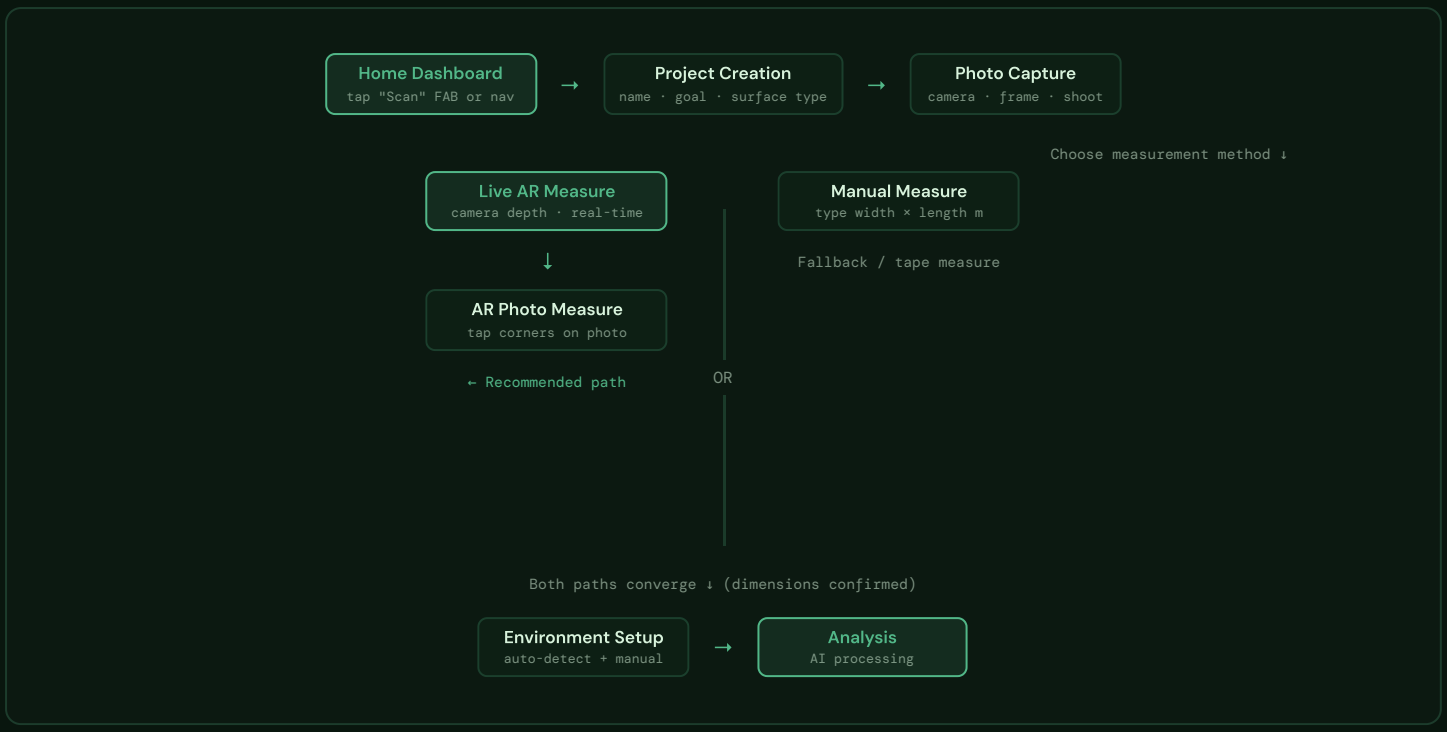
City / Area

Enter city name (min 2 chars)
Geolocation optional fallback
Proceeds on "Launch My Dashboard 🌿 "

SCREEN DETAIL — ONBOARDING SCREENS

SCREEN ID	SCREEN NAME	PURPOSE	NAVIGATES TO
splash	**Splash Screen**	Animated brand logo display on cold launch. Timed 3 s auto-advance or tap to skip.	→ onboarding
onboarding	**Onboarding Walkthrough**	3-card horizontal carousel explaining core value props: Scan → Plan → Cool. Swipe or tap arrows.	→ phone auth (last card "Get Started")
auth/phone	**Phone Authentication**	User enters mobile number. System sends OTP via SMS (Twilio). Dev mode logs OTP to console.	→ otp verify
auth/otp	**OTP Verification**	6-digit code entry. 30 s resend cooldown. Validates against hash+expiry in DB. Creates NextAuth session.	→ profileSetup (new user) → home (existing user)
profileSetup	**Profile Setup**	2-step form: name (step 1) and city (step 2). Saves user record. Defaults: experience=intermediate, role=homeowner.	→ home

JOURNEY 2 — FROM HOME TO MEASURED SPACE



SCREEN DETAIL — SCAN & MEASUREMENT SCREENS

SCREEN ID	SCREEN NAME	PURPOSE & KEY INTERACTIONS	NAVIGATES TO
create	**Project Creation**	User names the project, selects primary goal (cooling / mixed), sets surface type. Creates a PhotoSession record. Existing projects shown as cards.	→ photoCapture → existing project (tap)
photoCapture	**Photo Capture**	Full-screen camera with framing guide overlay. User taps shutter to capture JPEG. Option to retake. Stored as base64 data URL on photoSession.	→ liveARMeasure → photoMeasureAR → measure (manual)
liveARMeasure	**Live AR Measurement**	WebGL Three.js live view with depth-sensing. User points camera at space edges. Real-time distance readout. Tap to confirm corners. Derives widthM × lengthM + floorLevel.	→ environment (confirm)
photoMeasureAR	**AR Photo Measurement**	Shows captured photo. User taps 2 opposing corners on screen. Pixel distance + known reference → metric dimensions. Confidence score displayed.	→ environment (confirm)
measure	**Manual Measurement**	Numeric inputs for width (m) and length (m). Floor level selector. Optional tape-measure guide. Proceeds to environment when both fields filled.	→ environment
environment	**Environment Setup**	Auto-detects location, temperature, UV index, wind speed via geolocation + weather API. User can override sun exposure, wind level, space type sliders. Shows live weather card when detected.	→ analysis

ENVIRONMENT SCREEN — AUTO-DETECT SIGNALS

Location
GPS → lat/lng → city label

Temperature
Current + daily max °C

Wind Speed
km/h → sheltered/windy

UV Index
0–11 scale → heat zone

JOURNEY 3 — AI ANALYSIS TO GARDEN LAYOUT

AI RECOMMENDATION ENGINE — 3-TIER PIPELINE

Environment Screen (inputs locked) → **Analysis Screen** (POST /api/recommendations/generate)

TIER 1 · PRIMARY
ML Model
Python scoring process
Species feature vectors
Trained on telemetry

TIER 2 · FALLBACK
Catalog Hybrid
Rules + catalog merge
Pet-safe · drought filter
Heat-zone hard exclusions

TIER 3 · LAST RESORT
Rules Only
Deterministic rules
Always produces results
Lower confidence score

Analysis (results ready) → **Garden Layout** (architectural plan) → **Result Screen** (recommendation cards)

GARDEN LAYOUT SCREEN — PLAN CANVAS

Canvas Elements

- Architectural top-down plan (HiDPI canvas)
- Perimeter border beds (14% wide on all walls)
- 4 raised centre beds (A/B/C/D) in 2×2 grid
- Plant symbols: concentric rings (canopy/trunk)
- Compass rose, dimension lines with arrowheads
- Scale bar at bottom of canvas

Zone Legend & Actions

- PLANTING ZONES panel with coloured dots
- Species per zone with count badges
- Tap zone → expanded detail panel
- "View Detailed Recommendations →" CTA
- "View Full Report" secondary CTA

RESULT SCREEN — RECOMMENDATION CARDS

SCREEN ID	SCREEN NAME	PURPOSE & KEY INTERACTIONS	NAVIGATES TO
analysis	Analysis Screen	Shows animated progress while the 3-tier recommendation engine runs. Displays intermediate status (ML → hybrid → rules). Loading states with estimated °C reduction preview.	→ gardenLayout
gardenLayout	Garden Layout Screen	Renders architectural top-down canvas plan from layoutSchema. Shows zone labels, plant symbols, compass, scale. Swipeable zone detail panel at bottom.	→ result → report
result	Result Screen	Horizontally scrollable recommendation cards. Each card shows layout name, heat reduction (°C), cost range, top plants, confidence badge. Tap card → selected. Shows live AI-generated before/after thumbnail.	→ beforeAfter → report → install

RECOMMENDATION CARD — DATA FIELDS

Layout Name
Template or headline string

Heat Reduction
-X.X°C surface cooling

Cost Estimate
Materials + labour ₹ range

Confidence
high / medium / low badge

Coverage %
Plant / shade / reflective

Top Plants
Up to 3 species names

ROI
Payback months estimate

Modules
Count of cooling modules

JOURNEY 4 — FROM RESULT TO GARDEN VISUAL TO INSTALLATION

Result Screen (select recommendation) → **Before / After Visualization** (AI garden transformation)

THREE VIEW MODES WITHIN BEFORE/AFTER SCREEN:

SLIDER
Drag-to-reveal split
Before left / After right
Touch and mouse support

SIDE BY SIDE
2-column grid view
Before left / After right
190px thumbnail height

LAYOUT OVERLAY
Photo + zone canvas
Perspective-mapped anchors
Coloured zone footprints

Actions from Before/After screen ↓

- **Report Screen** — full PDF report
- **Install Screen** — request quote
- **Regenerate** — new AI visual

AI VISUALIZATION — GPT-IMAGE-1 PIPELINE

Input Signals to Prompt

- Space dimensions (widthM × lengthM m²)
- Floor level → elevation context
- Space type (rooftop / balcony / terrace)
- Environment (sun exposure, wind, temp °C)
- Scored plants (up to 12): type + visual descriptor
- Layout zones: positions as % of total space
- Infrastructure: planter module positions
- Heat reduction targets (coverage%, shade%, °C)

Output & Storage

- gpt-image-1 → 1024×1024 PNG (base64)
- Stored as VisualizationRecord in DB
- SourcePhotoRef (128 char ref)
- LayoutSchema + SpatialMapping JSON
- VisualizationPrompt (2000 chars)
- GenerationVersion counter per session
- Displayed auto on screen entry (once)
- Manual "Regenerate" triggers new call

REPORT & INSTALLATION SCREENS

SCREEN ID	SCREEN NAME	PURPOSE & KEY INTERACTIONS	NAVIGATES TO
beforeAfter	**Before/After Visualization**	Auto-generates AI garden photo on entry. Three modes: Slider (drag-reveal), Side-by-Side (grid), Layout Overlay (canvas zone markers on photo). Shows recommendation summary, cost, modules, plants, confidence.	→ report → install → result (back)
report	**Report Screen**	Full recommendation report. Heat reduction breakdown, zone-by-zone plant list, cost breakdown (materials + labour + maintenance), ROI timeline chart, installation notes. Download PDF or share.	→ install (Get Quote) → result (back)
install	**Install Screen**	Request installation quote form. User selects preferred timing, budget flexibility, notes. Assigns nearest installer from system pool. Sends notification to installer.	→ installDone
installDone	**Install Success Screen**	Confirmation screen with installer contact details, expected timeline, next steps. Option to view impact dashboard or return to home.	→ home → impact

JOURNEY 5 — EXPLORE & DISCOVERY (CITY HEAT TAB)

City Heat Screen (heatmap · explore tab) → Species Library (browse all plants) → Species Detail (deep plant info) → Create / Scan ("Include in New Scan →")

SCREEN ID	SCREEN NAME	PURPOSE & KEY INTERACTIONS	NAVIGATES TO
cityHeat	City Heat Screen	City-scale heat map overlay. Shows urban heat island zones, rooftop temperature readings, and neighbourhood cooling potential. Filter by zone type. Entry point to species library.	→ speciesLib → create
speciesLib	Species Library	Searchable grid of all cooling plants in catalog. Filter by type (herb, climber, succulent…), pet-safe, drought-tolerant, sun requirement. Tap card → detail.	→ speciesDetail → cityHeat (back)
speciesDetail	Species Detail Screen	Full plant profile: photo, scientific name, heat reduction index, water needs, maintenance level, pet-safe status, drought tolerance, seasonal behaviour, planting tips.	→ create (CTA) → speciesLib (back)

JOURNEY 6 — IMPACT DASHBOARD

Impact Dashboard (leaf tab) → Carbon Dashboard (emissions tracker) → Carbon Setup (configure baseline)

SCREEN ID	SCREEN NAME	PURPOSE & KEY INTERACTIONS	NAVIGATES TO
impact	Impact Dashboard	Cumulative cooling impact across all user projects. Shows total area greened (m^2), estimated °C reduction, kg CO_2 sequestered, water saved (L). Animated stat cards. Gamification badges.	→ carbon → create
carbon	Carbon Dashboard	Monthly carbon offset chart. Current vs baseline emissions. Breakdown by project. Shows trees-equivalent and energy-saved kWh metrics. Links to setup if baseline not configured.	→ carbonSetup → home (back)
carbonSetup	Carbon Setup	User inputs electricity consumption (kWh/month), AC usage hours/day, household size. System calculates baseline carbon footprint and cooling offset potential.	→ carbon (save)

JOURNEY 7 — UTILITY & SETTINGS SCREENS

SCREEN ID	SCREEN NAME	PURPOSE & KEY INTERACTIONS	NAVIGATES TO
settings	Settings Screen	Account info, notifications toggle, unit preferences (metric/imperial), connected accounts, privacy, sign out. Edit profile link.	→ home → phone auth (sign out)
saved	Saved Projects Screen	List of all completed photo sessions and their recommendations. Tap to resume or view report. Sort by date or heat reduction.	→ result (tap project) → create
notifications	Notifications Screen	System notifications: new installer quote, seasonal plant tips, weather alerts for user's region, badge unlocks. Mark as read / clear all.	→ home → install (quote notif)
tips	Tips Screen	Curated gardening and cooling tips. Personalised to user's space type and region. Expandable tip cards with action buttons.	→ home → speciesLib

ALL 26 SCREENS — CATEGORISED

ONBOARDING First-time & auth screens

`splash`
Splash Screen
Animated logo · auto-dismiss

`onboarding`
Onboarding Walkthrough
3-card value prop carousel

`profileSetup`
Profile Setup
Name + city · 2-step form

CORE FLOW Primary scan-to-recommendation pipeline

`home`
Home Dashboard
Activity feed · quick CTA · carbon strip

`create`
Project Creation
Name · goal · existing projects

`photoCapture`
Photo Capture
Camera · frame guide · retake

`liveARMeasure`
Live AR Measure
Three.js depth · real-time dims

`photoMeasureAR`
AR Photo Measure
Tap corners on captured photo

`measure`
Manual Measurement
Numeric width × length inputs

`environment`
Environment Setup
Auto-detect + manual override

`analysis`
Analysis Screen
3-tier AI engine · progress UI

`gardenLayout`
Garden Layout
Architectural plan canvas · zones

`result`
Result Screen
Recommendation cards · select

VISUAL & ACTION Post-recommendation screens

`beforeAfter`
Before / After
AI visual · slider · overlay modes

`report`
Report Screen
Full PDF · cost · ROI · zones

`install`
Install Screen
Quote request · installer assign

`installDone`
Install Success
Confirmation · contact details

EXPLORE Discovery & species catalog

`cityHeat`
City Heat Screen
Urban heat map · zone explorer

`speciesLib`
Species Library
Searchable plant catalog grid

`speciesDetail`
Species Detail
Full plant profile · traits · tips

IMPACT & UTILITY Impact tracking, settings, support

`impact`
Impact Dashboard
Cumulative cooling · CO_2 · badges

`carbon`
Carbon Dashboard
Monthly offset chart · kWh saved

`carbonSetup`
Carbon Setup
Electricity baseline · household

`saved`
Saved Projects
Past scans · resume · sort

`notifications`
Notifications
Alerts · quotes · tips · badges

`tips`
Tips Screen
Personalised gardening tips

`settings`
Settings
Account · prefs · sign out